

Youwei Xiao

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Education

Bachelor of Science degree in Computer Science

September 2018 - July 2022

School of Electronics Engineering and Computer Science, Peking University

Beijing, China

Ph.D. Candidate in EDA

September 2022 - now

School of Integrated Circuits, Peking University

Beijing, China

Research

My research centers on compiler techniques for MLSys, computer architecture, and EDA, with emphasis on domain-specific languages, intermediate representations, and program optimization.

Hardware Synthesis & EDA: I develop compiler infrastructures, multi-level intermediate representations, and synthesis methods that bridge high-level specifications and RTL implementations. Key contributions include *Cement* (FPGA '24), a cycle-deterministic embedded HDL in Rust, and *Hector* (ICCAD '22), an MLIR-based multi-level IR for hardware synthesis. I also explore e-graph techniques for hardware optimization in the *SkyEgg* project.

Computer Architecture: I work on compiler-guided generation of domain-specific accelerators and reusable custom instructions. *Cayman* (DAC '25) combines application profiling with design space exploration for accelerator generation. *ISAMORE* (ASPLOS '26 Best Paper Award, 5/1048) uses e-graph anti-unification to find reusable custom instructions.

Hardware-Software Co-Design: I lead the *APS* project (ICCAD '25 Invited), a compiler-centered framework that generates optimized processor architectures, RTL implementations, and compiler support from target applications—towards fully automated co-design without manual intervention.

ML Compilers & Systems: I am actively researching compiler-centric ML systems including human- and LLM-friendly retargetable tensor compilers (*HTP*), task-graph runtimes for distributed tensor execution (*PTO Runtime*), execution-guided superoptimization (*EggMind*), and explicit orchestration substrates for multi-agent systems (*Hive*).

Publications

- **Youwei Xiao**, Chenyun Yin, Yitian Sun, and Yun Liang. 2026. Finding Reusable Instructions via E-Graph Anti-Unification. In Proceedings of the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '26). **[Best Paper Award, 5/1048]**
- **Youwei Xiao**, Yuyang Zou, Yansong Xu, Yuhao Luo, Yitian Sun, Chenyun Yin, Ruifan Xu, Renze Chen, and Yun Liang. 2025. APS: Open-Source Hardware-Software Co-Design Framework for Agile Processor Specialization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). [Invited Paper]
- Weijie Peng*, **Youwei Xiao***, Yuyang Zou, Zizhang Luo, and Yun Liang. 2025. Clay: High-level ASIP Framework for Flexible Microarchitecture-Aware Instruction Customization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). (*Equal contribution)
- **Youwei Xiao**, Fan Cui, Zizhang Luo, Weijie Peng, and Yun Liang. 2025. Cayman: Custom Accelerator Generation with Control Flow and Data Access Optimization. In Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC '25).
- **Youwei Xiao**, Zizhang Luo, and Yun Liang. 2025. cmt2: Rule-Based Hardware Description in Rust with Temporal Semantics. In 5th Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE '25).
- Fan Cui, **Youwei Xiao**, Kexing Zhou, and Yun Liang. 2025. An Empirical Comparison of LLM-based Hardware Design and High-level Synthesis. In Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '25).

- Fan Cui, Chenyang Yin, Kexing Zhou, **Youwei Xiao**, Guangyu Sun, Qiang Xu, Qipeng Guo, Yun Liang, Xingcheng Zhang, Demin Song, and Dahua Lin. 2024. OriGen: Enhancing RTL Code Generation with Code-to-Code Augmentation and Self-Reflection. In Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD '24).
- **Youwei Xiao**, Zizhang Luo, Kexing Zhou, and Yun Liang. 2024. Cement: Streamlining FPGA Hardware Design with Cycle-Deterministic eHDL and Synthesis. In Proceedings of the 2024 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '24).
- Ruifan Xu, **Youwei Xiao**, Jin Luo, and Yun Liang. 2022. HECTOR: A Multi-Level Intermediate Representation for Hardware Synthesis Methodologies. In Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design (ICCAD '22).

Preprints

- Yuyang Zou, **Youwei Xiao**, Yansong Xu, Chenyun Yin, Yuhao Luo, Yitian Sun, Ruifan Xu, Renze Chen, and Yun Liang. Aquas: Enhancing Domain Specialization through Holistic Hardware-Software Co-Optimization based on MLIR. arXiv:2511.22267.
- **Youwei Xiao**, Yuyang Zou, and Yun Liang. SkyEgg: Joint Implementation Selection and Scheduling for Hardware Synthesis using E-graphs. arXiv:2511.15323.
- **Youwei Xiao**, Zizhang Luo, Weijie Peng, Yuyang Zou, and Yun Liang. Cement2: Temporal Hardware Transactions for High-Level and Efficient FPGA Programming. arXiv:2511.15073.

Honors and Awards

ASPLOS 2026 Best Paper Award	March 2026
Finding Reusable Instructions via E-Graph Anti-Unification	5/1048 submissions
Second Place in EDathon 2020	August 2020
Programming Competition on Electronic Design Automation	IEEE CEDA Hong Kong Chapter
Shenzhen Stock Exchange Scholarship	2019-2020
Awarded to 17 students from EECS, Peking University	
Merit Student	2018-2019 & 2019-2020
Awarded to top 10% students for comprehensive excellence	
Yang Fuqing & Wang Yangyuan Academician Scholarship	2018-2019
Awarded to only 5 students at EECS, Peking University, for academic excellence	
Second Prize in NOI 2017	July 2017
National Olympiad in Informatics	China Computer Federation
Second Prize in APIO 2017	May 2017
Asia Pacific Informatics Olympiad (China District)	China Computer Federation

Tutorials

APS: An MLIR-Based Hardware-Software Co-design Framework	January 2026
ASP-DAC 2026	Tokyo, Japan
Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design	May 2025
ISED 2025	Xi'an, China
Agile Hardware Specialization (AHS)	March 2025
ASPLOS 2025	Rotterdam, Netherlands
Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design	March 2025
DATE 2025	Lyon, France
AHS: An EDA toolbox for Agile Chip Front-end Design	January 2025
ASP-DAC 2025	Tokyo, Japan

Projects and Research

APS: Agile Processor Specialization	2023 – Present
Open-source framework for automated processor specialization from applications to RTL	GitHub
Cement: Cycle-Deterministic eHDL	2022 – Present
Embedded HDL in Rust with temporal transactions for FPGA design	GitHub
Hector: Multi-Level Hardware IR	2021 – 2022
MLIR-based multi-level intermediate representation for hardware synthesis	GitHub
ISAMORE: Instruction Reuse via E-Graphs	2025 – Present
E-graph anti-unification for reusable custom instructions; ASPLOS 2026 Best Paper Award (5/1048)	
Cayman: Custom Accelerator Generation	2024 – 2025
Accelerator generation with control flow and data access optimization	
SkyEgg: Hardware Synthesis with E-Graphs	2024 – Present
Joint implementation selection and scheduling using e-graphs	
HTP	2025 – Present
Human- and LLM-friendly retargetable compiler substrate with inspectable IRs	GitHub
PTO Runtime	2025 – Present
Task-graph runtime for distributed tensor compilation and host-device execution	GitHub
EggMind: Agentic Superoptimizer	2025 – Present
Execution-guided agentic superoptimizer with bounded search for scalable equality saturation	OpenReview
Hive	2025 – Present
Explicit orchestration substrate for programmable multi-agent systems	GitHub

Skills

Programming Languages: Rust, C++, Python, Scala, Verilog, CUDA
Compiler Infrastructure: MLIR, LLVM
ML Frameworks: PyTorch, Tilelang, Triton